

Bootstrap - Page 1

Introduction to Bootstrap

Bootstrap is a popular CSS framework used to build responsive and mobile-first websites.

Features of Bootstrap

- Responsive grid system
- Pre-designed components
- Consistent styling
- Easy customization
- Cross-browser compatibility

Ilovecoding

Developed By

Mark Otto & Jacob Thornton

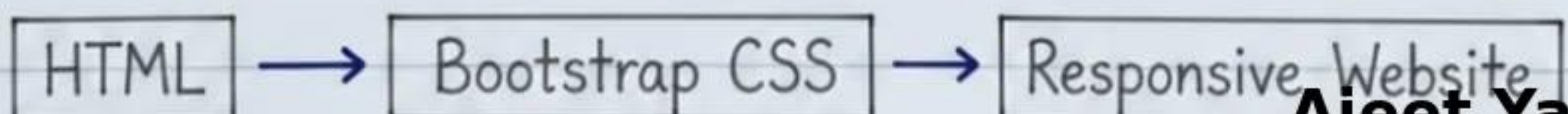
Basic Bootstrap Example

```
<link rel="stylesheet"
href="https://cdn.jsdelivr.net/npm/
bootstrap@5.3.0/dist/css/bootstrap.min.css">
```

Components in Bootstrap

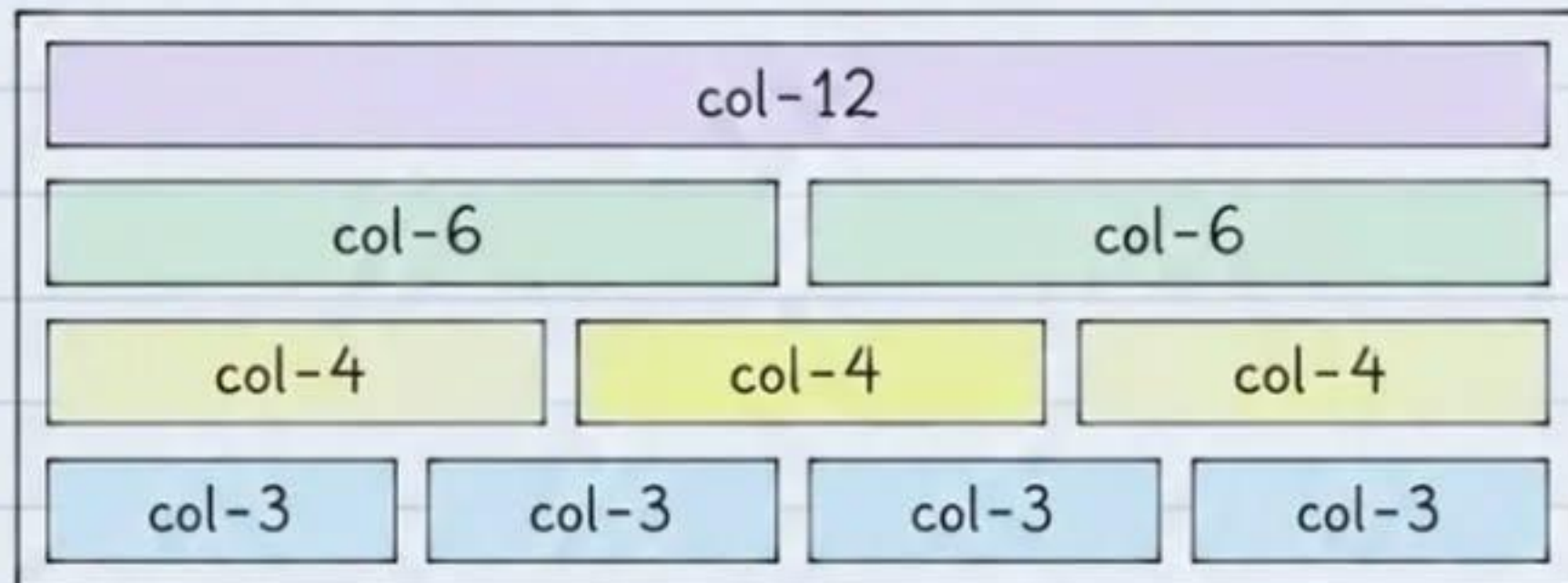
- Buttons
- Navigation Bar
- Cards
- Forms
- Modals

Exam Tip :
Bootstrap saves time by providing ready-to-use components.



Grid System

Bootstrap uses a 12-column grid system to create responsive layouts.



Grid Breakpoints

- xs : Extra small devices (< 576px)
- sm : Small devices (≥ 576px)
- md : Medium devices (≥ 768px)
- lg : Large devices (≥ 992px)
- xl : Extra large devices (≥ 1200px)
- xxl : Extra extra large devices (≥ 1400px)

Exam Tip :
Understand breakpoints to build truly responsive designs.

Utility Classes

Bootstrap provides many utility classes for spacing, colors, text, borders, and more.

Category	Examples
Spacing	m-3, p-4, mt-2, mb-3
Text	text-center, text-primary, fw-bold
Colors	bg-primary, text-success, border-danger
Display	d-none, d-block, d-flex
Flexbox	justify-content-center, align-items-center

Exam Tip :
Utility classes help you write less CSS and more HTML.

Including Bootstrap

Bootstrap can be included in two ways:

- **CDN** : `<link rel="stylesheet" href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css">`
- **Local** : Download Bootstrap and link the CSS and JS files.

Container vs Container-Fluid

Bootstrap provides two container classes for layout.

Class	Description
.container	Fixed width container that changes at each breakpoint.
.container-fluid	Full width container that spans the entire width of the viewport.

Exam Tip :

Use .container for fixed layouts and .container-fluid for full width layouts.

Typography

Bootstrap provides pre-styled typography classes.

- Display Headings : .display-1 to .display-6
- Headings : .h1 to .h6
- Lead Text : .lead
- Muted Text : .text-muted
- Alignment : .text-start, .text-center, .text-end
- Text Transformation : .text-uppercase, .text-lowercase, .text-capitalize

Exam Tip :

Typography classes ensure consistency and improve readability.

Buttons

Bootstrap provides different styles and colors for buttons.

```
<button type="button" class="btn btn-primary"> Primary</button>
<button type="button" class="btn btn-success"> Success</button>
<button type="button" class="btn btn-danger"> Danger</button>
<button type="button" class="btn btn-outline-primary"> Outline</button>
```

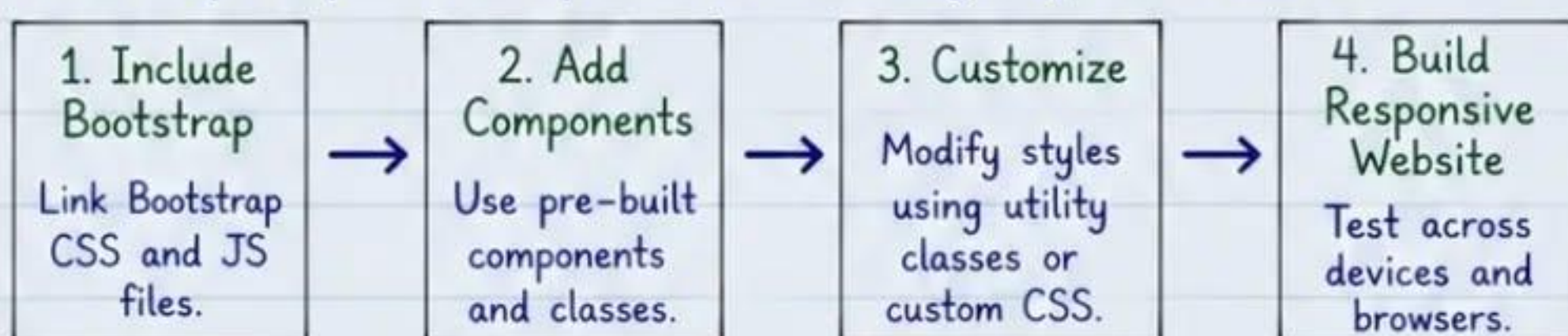


Exam Tip :

Use button classes to create visually appealing and consistent buttons.

Bootstrap Workflow

Bootstrap simplifies the process of building responsive websites.



★ Bootstrap helps you build responsive, modern, and consistent websites faster!

Responsive Breakpoints

Bootstrap is mobile-first and uses the following breakpoints.

Breakpoint	Class Infix	Min Width	Target Devices
Extra small	None	< 576px	Phones
Small	sm	≥ 576px	Small tablets
Medium	md	≥ 768px	Tablets
Large	lg	≥ 992px	Desktops
Extra large	xl	≥ 1200px	Large desktops
Extra extra large	xxl	≥ 1400px	Extra large desktops

Exam Tip :
Bootstrap is mobile-first. Styles apply from the smallest screen upwards.

Grid System Example

Example of a simple 12-column layout.

```
<div class="container">
  <div class="row">
    <div class="col-md-4">Column 1</div>
    <div class="col-md-4">Column 2</div>
    <div class="col-md-4">Column 3</div>
  </div>
</div>
```

Exam Tip :
Use rows to wrap columns. Total columns in a row should add up to 12.

Images

Bootstrap provides responsive image classes.

Class	Description
.img-fluid	Makes images responsive. Scales with the parent element.
.img-thumbnail	Adds padding and a border around the image.

Exam Tip :
Use .img-fluid for responsive images.

Alerts

Bootstrap provides contextual alert messages.

```
<div class="alert alert-success" role="alert">
  Success! This is a success alert.
</div>
<div class="alert alert-danger" role="alert">
  Error! This is a danger alert.
</div>
<div class="alert alert-warning" role="alert">
  Warning! This is a warning alert.
</div>
<div class="alert alert-info" role="alert">
  Info! This is an info alert.
</div>
```

Exam Tip :
Use alerts to display important messages to users.

Tables

Bootstrap provides classes to style tables.

Class	Description
.table	Basic table with light padding and horizontal dividers.
.table-striped	Adds zebra-stripping to rows.
.table-bordered	Adds borders on all sides of the table and cells.
.table-hover	Highlights table rows on hover.
.table-responsive	Makes the table responsive by adding a horizontal scroll on small screens.

Exam Tip :
Use table classes to improve readability and responsiveness.

Forms

Bootstrap provides classes to style and layout forms.

```
<form>
  <div class="mb-3">
    <label for="name" class="form-label">Name</label>
    <input type="text" class="form-control" id="name">
  </div>
  <div class="mb-3">
    <label for="email" class="form-label">Email</label>
    <input type="email" class="form-control" id="email">
  </div>
  <div class="form-check mb-3">
    <input class="form-check-input" type="checkbox" id="agree">
    <label class="form-check-label" for="agree">
      I agree to the terms and conditions
    </label>
  </div>
  <button type="submit" class="btn btn-primary">Submit</button>
</form>
```

Exam Tip :
Use form-control for inputs, form-label for labels, and form-check for checkboxes and radios.

Cards

Bootstrap provides card component for flexible and extensible content.



```
<div class="card" style="width: 18rem;">
  
  <div class="card-body">
    <h5 class="card-title">Card Title</h5>
    <p class="card-text">
      This is some example text inside the card body.
    </p>
    <a href="#" class="btn btn-primary">Read More</a>
  </div>
</div>
```

Exam Tip :
Cards are great for displaying content in a clean and organized manner.

Conclusion

Bootstrap is a powerful front-end framework that helps you build responsive, mobile-first websites quickly and efficiently using pre-built components and utilities.

★ Learn Bootstrap, build faster, and make your websites look amazing!

Ajeet Yadav

Navigation Bar

Bootstrap provides classes for creating responsive navigation bars.

```
<nav class="navbar navbar-expand-lg navbar-light bg-light">
  <div class="container-fluid">
    <a class="navbar-brand" href="#">Brand</a>
    <button class="navbar-toggler" type="button" data-bs-toggle="collapse"
      data-bs-target="#navbarNav" aria-controls="navbarNav"
      aria-expanded="false" aria-label="Toggle navigation">
      <span class="navbar-toggler-icon"></span>
    </button>
    <div class="collapse navbar-collapse" id="navbarNav">
      <ul class="navbar-nav ms-auto">
        <li class="nav-item"><a class="nav-link active" href="#">Home</a></li>
        <li class="nav-item"><a class="nav-link" href="#"> About</a></li>
        <li class="nav-item"><a class="nav-link" href="#"> Contact</a></li>
      </ul>
    </div>
  </div>
</nav>
```

Exam Tip :
Use navbar classes and .navbar-toggler for responsive navigation menus.

Modals

Bootstrap modals are used to display content in a popup window.

Open Modal

```
<div class="modal fade" id="exampleModal" tabindex="-1">
  <div class="modal-dialog">
    <div class="modal-content">
      <div class="modal-header">
        <h5 class="modal-title"> Modal Title</h5>
        <button type="button" class="btn-close" data-bs-dismiss="modal"></button>
      </div>
      <div class="modal-body">
        This is some modal content.
      </div>
      <div class="modal-footer">
        <button type="button" class="btn btn-secondary" data-bs-dismiss="modal">Close</button>
        <button type="button" class="btn btn-primary">Save changes</button>
      </div>
    </div>
  </div>
</div>
```

Exam Tip :
Use modals to display important content without navigating away from the page.

Collapse

Bootstrap collapse is used to show or hide content.

```
<p>
  <a class="btn btn-primary" data-bs-toggle="collapse"
    href="#collapseExample" role="button" aria-expanded="false"
    aria-controls="collapseExample">
    Toggle content
  </a>
</p>
<div class="collapse" id="collapseExample">
  <div class="card card-body">
    This content is hidden by default
    and shown when the button is clicked.
  </div>
</div>
```

Exam Tip :
Use collapse to show or hide content for FAQs, accordions, and more.

Tooltips & Popovers

Bootstrap provides tooltips and popovers to show additional information.

- Tooltip: Small text on hover.
- Popover: More content on click or hover.

```
<button type="button" class="btn btn-info"
  data-bs-toggle="tooltip" data-bs-placement="top"
  title="Tooltip example">
  Hover me
</button>
```

Exam Tip :
Tooltips and popovers improve user experience.

